

Data Visualization Essentials: Techniques, Accessibility, and Comparisons

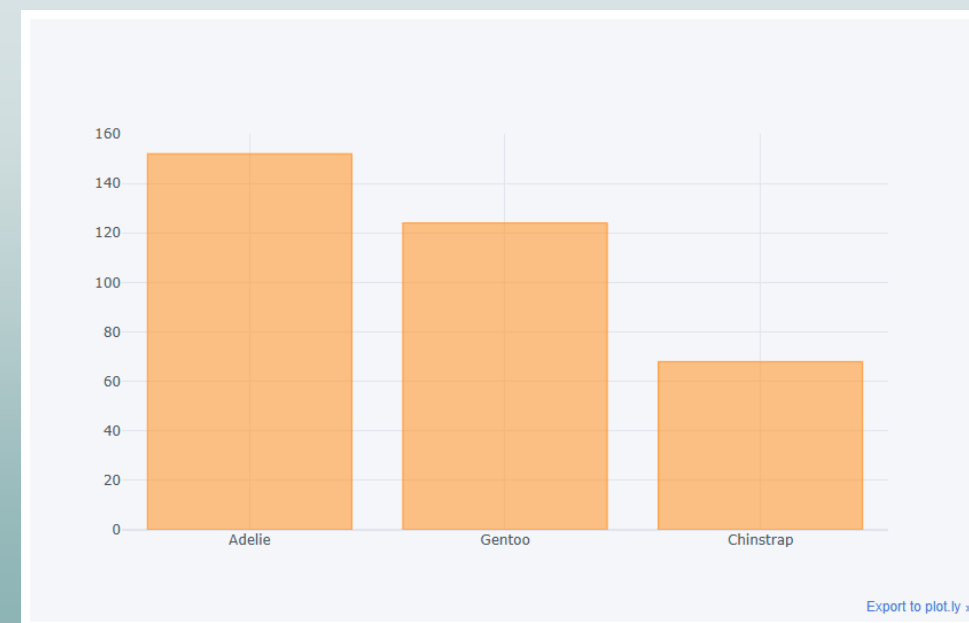
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Why should I use visualizations to represent my data?, What are some visualizations that I can use to represent my data?, How can I make my visualizations be more accessible?, are there any visualizations that are similar but where they are used can vary based on the data?, Is it possible to use a wrong visualization for your dataset?

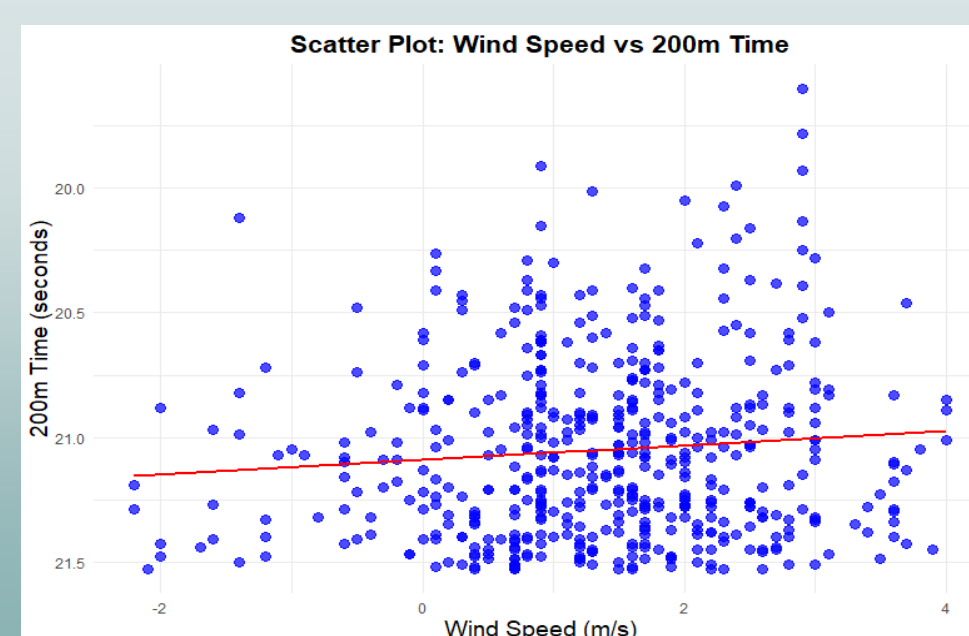
Why Visualize?

Visualizing data is essential because it transforms complex datasets into understandable insights, allowing stakeholders to quickly grasp key trends and patterns. In a data-rich environment, effective visualizations, such as charts and graphs, reveal relationships that may not be obvious in raw data, fostering a deeper understanding and enabling informed decision-making. Additionally, visual representations enhance engagement and retention, as people tend to remember visual information better than text or numbers. By providing a common framework for analysis, data visualization also promotes collaboration and productive discussions among team members. Ultimately, it is a powerful tool that enhances comprehension and facilitates effective communication in a data-driven world.

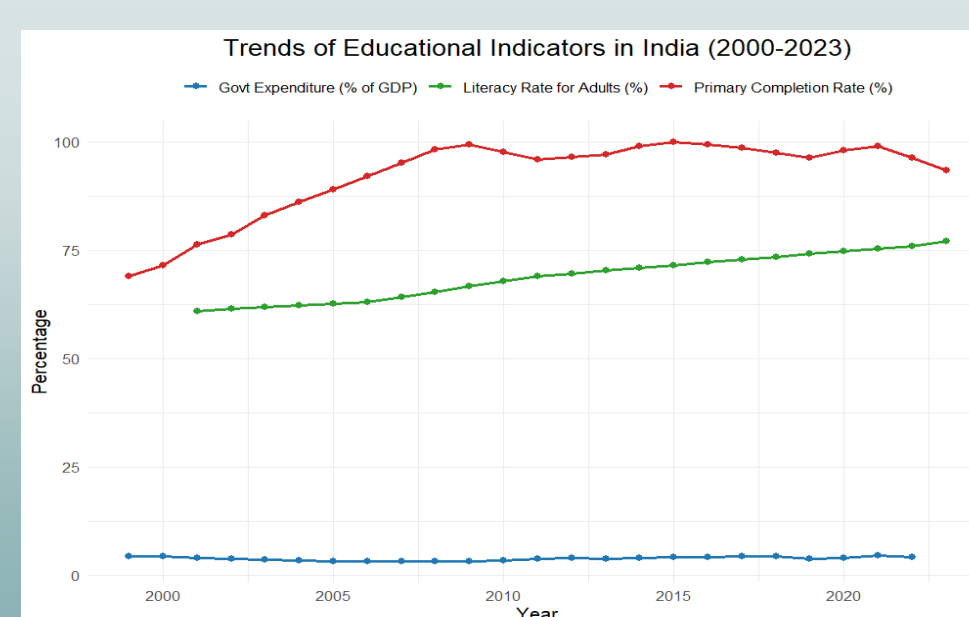
Different Visualizations



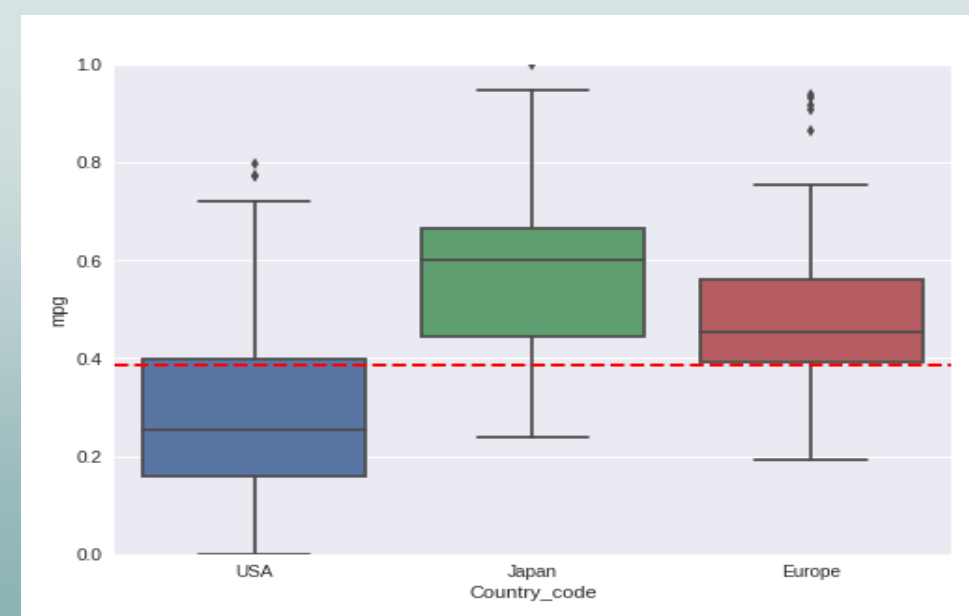
- **Benefits:**
 - Easy to read and interpret.
 - Effective for comparing quantities across categories.
- **Use Cases:**
 - Count of all the categories in a feature.



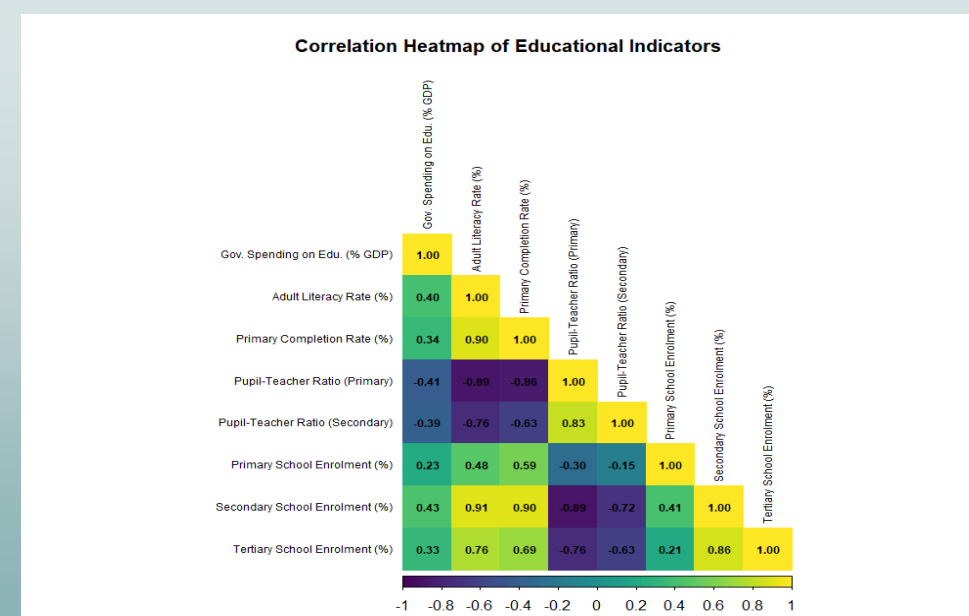
- **Benefits:**
 - Reveals relationships and correlations between two variables.
 - Helps identify outliers and clusters in data.
- **Use Cases:**
 - Analyzing correlation between Wind Speed and Time.



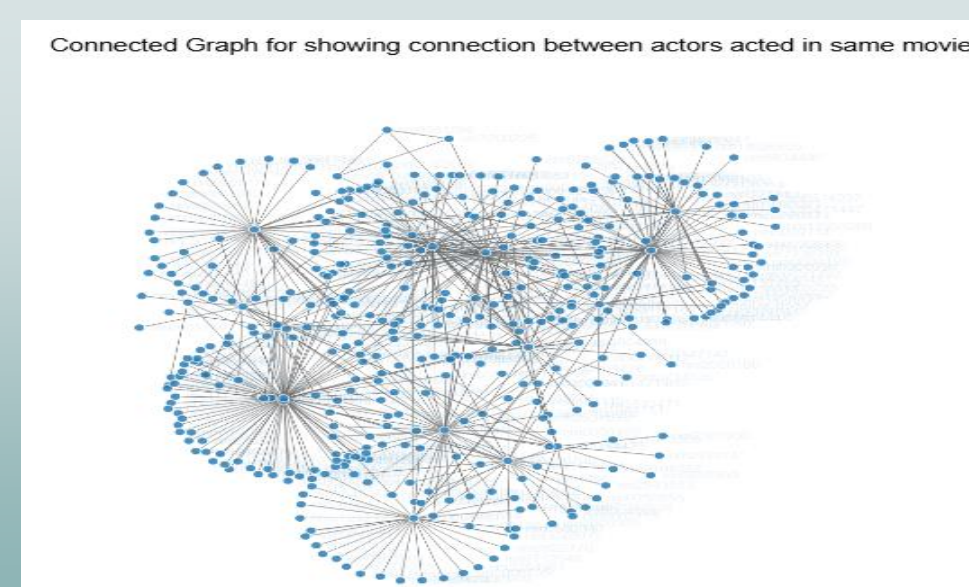
- **Benefits:**
 - Ideal for displaying trends over time.
 - Highlights changes and patterns clearly.
- **Use Cases:**
 - Analyzing Trends over time.



- **Benefits:**
 - Summarizes data through quartiles, highlighting median and outliers.
 - Useful for comparing distributions across different groups.
- **Use Cases:**
 - Comparing test scores across different classes or schools.

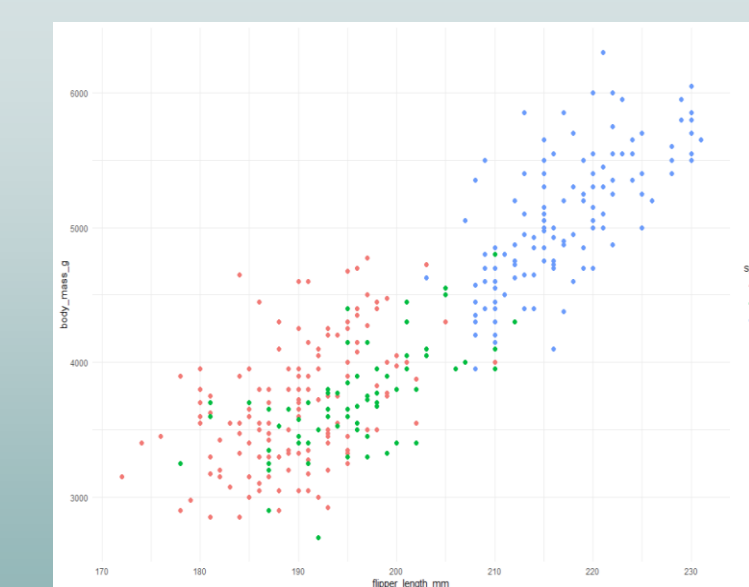


- **Benefits:**
 - Represents data density or intensity through color gradients.
 - Effective for visualizing complex data sets in a compact format.
- **Use Cases:**
 - Analyzing correlations between different features

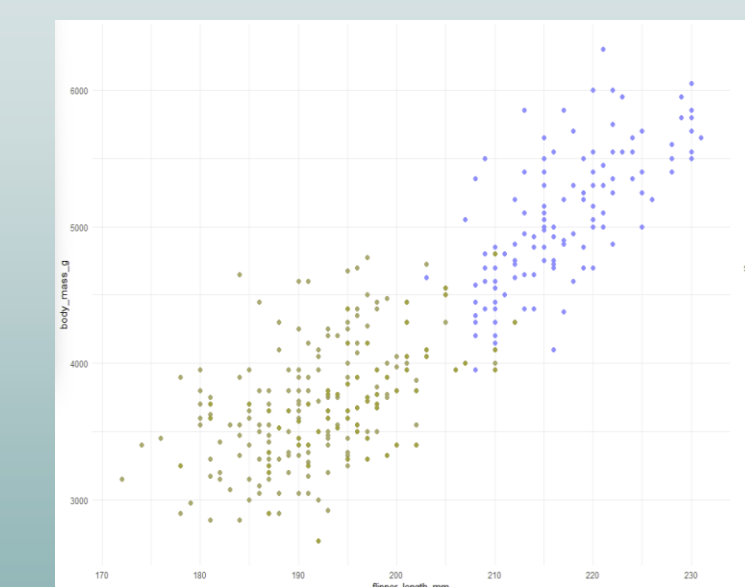


- **Benefits:**
 - Visualize complex relationships and interactions between nodes, making it easier to identify patterns and clusters within the data.
- **Use Cases:**
 - Social Network Analysis

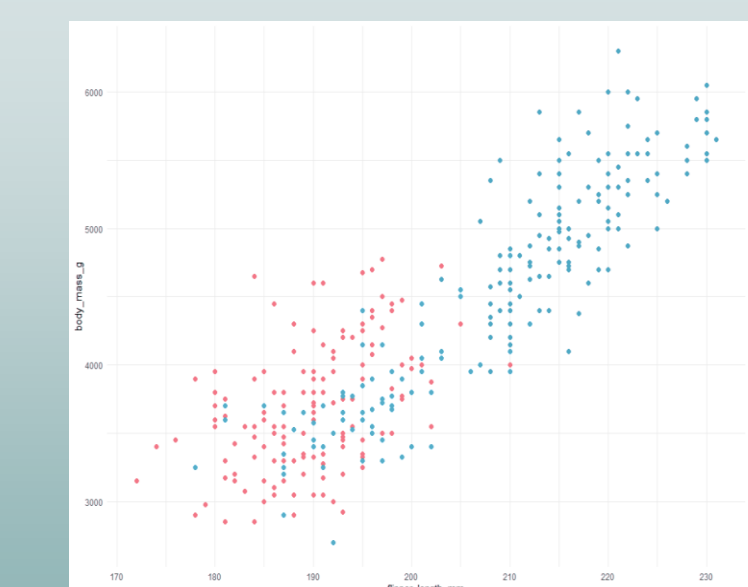
Accessibility



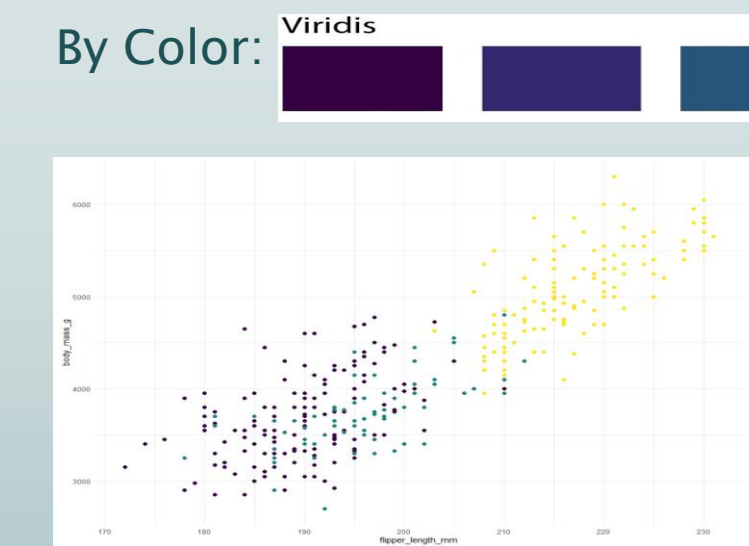
Original



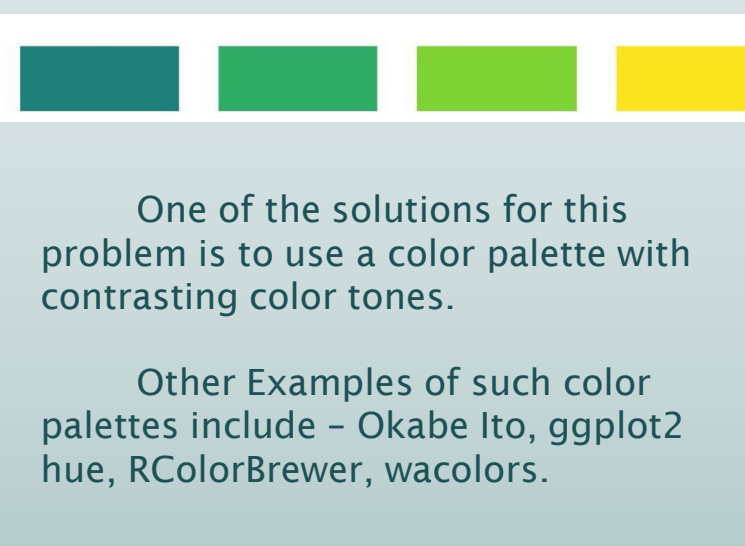
Deuteranopia(Green-Blind)



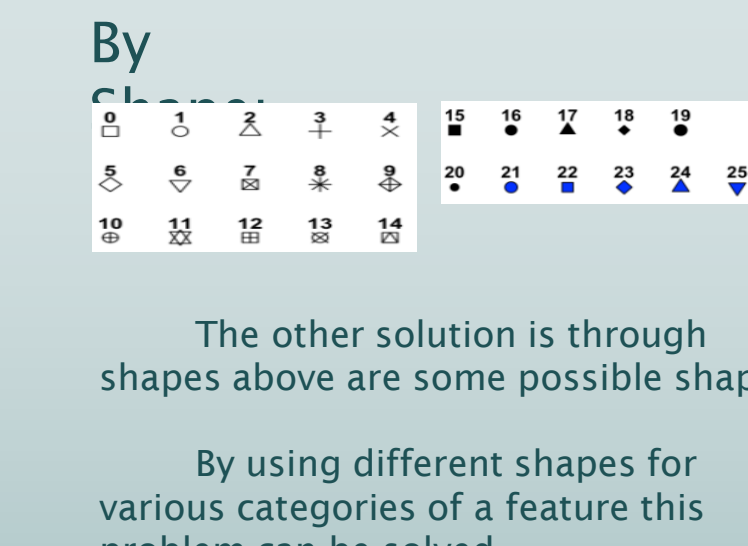
Tritanopia(Blue-Blind)



Solution



Deuteranopia(Green-Blind)



Tritanopia(Blue-Blind)

Increasing accessibility by shape

The Problem

By Color: Viridis

One of the solutions for this problem is to use a color palette with contrasting color tones.

Other Examples of such color palettes include - Okabe Ito, ggplot2 hue, RColorBrewer, wacolors.

The Solution

By

The other solution is through shapes above are some possible shapes.

By using different shapes for various categories of a feature this problem can be solved.

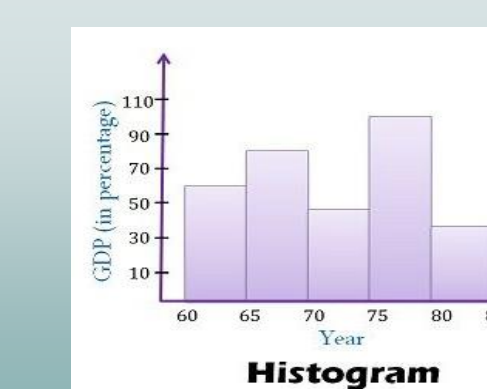
Comparisons between Visualizations



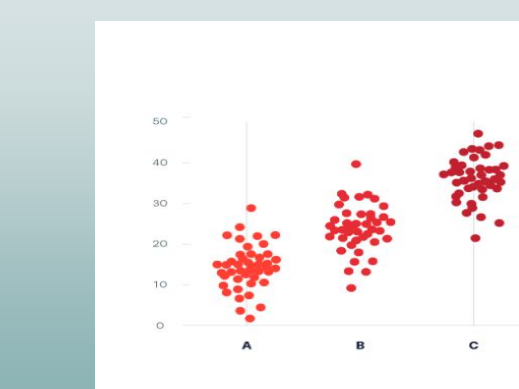
Bar Graph

Bar Graph vs Histogram

- Bar Graph, While these two visualizations are similar Bar Graph is best used for feature with discrete/ categorical values
- While Histogram is best used for feature with continuous values.

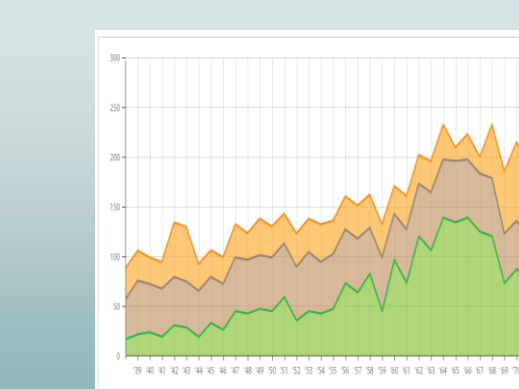
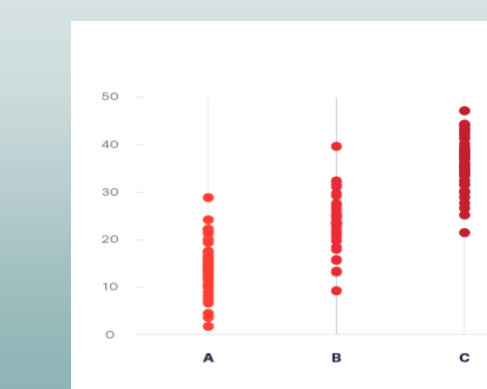


Histogram



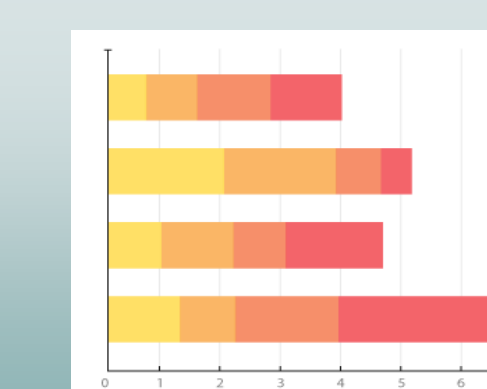
Jitter Plot vs Strip Plot

- Jitter Plot, It include displacing the datapoint so that there will be no overlapping and therefore be quantified.
- As for the Strip Plot, Its straightforward with no displacement complexity.



Stacked Area Chart vs Stacked Bar Chart

- Stacked Area Chart is useful for studying overall.
- As for the Stacked Bar Chart, Its best used for comparison among parts.



References:

- <https://www.kaggle.com/code/devanshbesain/exploration-and-analysis-auto-mpg>
- <https://allisonhorst.github.io/palmerpenguins/articles/intro.html>
- <https://datavizproject.com/data-type/strip-plot/#~:text=A%20strip%20plot%20is%20a,the%20same%20value%20can%20overlap.>
- https://keydifferences.com/difference-between-histogram-and-bar-graph.html#google_vignette
- <https://datavizproject.com/data-type/jitter-plot/>
- <https://datavizproject.com/data-type/strip-plot/>
- https://datavizcatalogue.com/methods/stacked_bar_graph.html
- https://datavizcatalogue.com/methods/stacked_area_graph.html

Why Access matters?

Making data visualizations more accessible is essential for ensuring that everyone, regardless of their abilities or backgrounds, can effectively engage with the information. Accessible visuals allow individuals with disabilities, such as visual impairments, to understand data through screen readers or descriptive text. Simplifying complex visuals and using clear language also help those without a strong background in data analysis. By prioritizing accessibility, organizations promote inclusivity and encourage broader participation in discussions and decision-making. Ultimately, accessible visualizations enhance the reach and impact of data, making valuable insights available to all.

Why this section of comparing?

This section is to show that context about what is expected of the visualization and the type of data that's used for the visualization matters in the visualization chosen.

In this section we are giving three examples where these context make a difference.